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Introduction to Data Communications

Topic 02 – *Application Layer*

Our vision is to be Singapore's private education institution of choice, and we are committed to helping our students achieve their career and educational goals through lifelong learning opportunities.

We build futures, one success story at a time.

Integrity . Knowledge . Support . Opportunity . Results



Overview

Network Application Architectures

- -Client-Server
- -Peer-to-peer

Application Layer Protocols

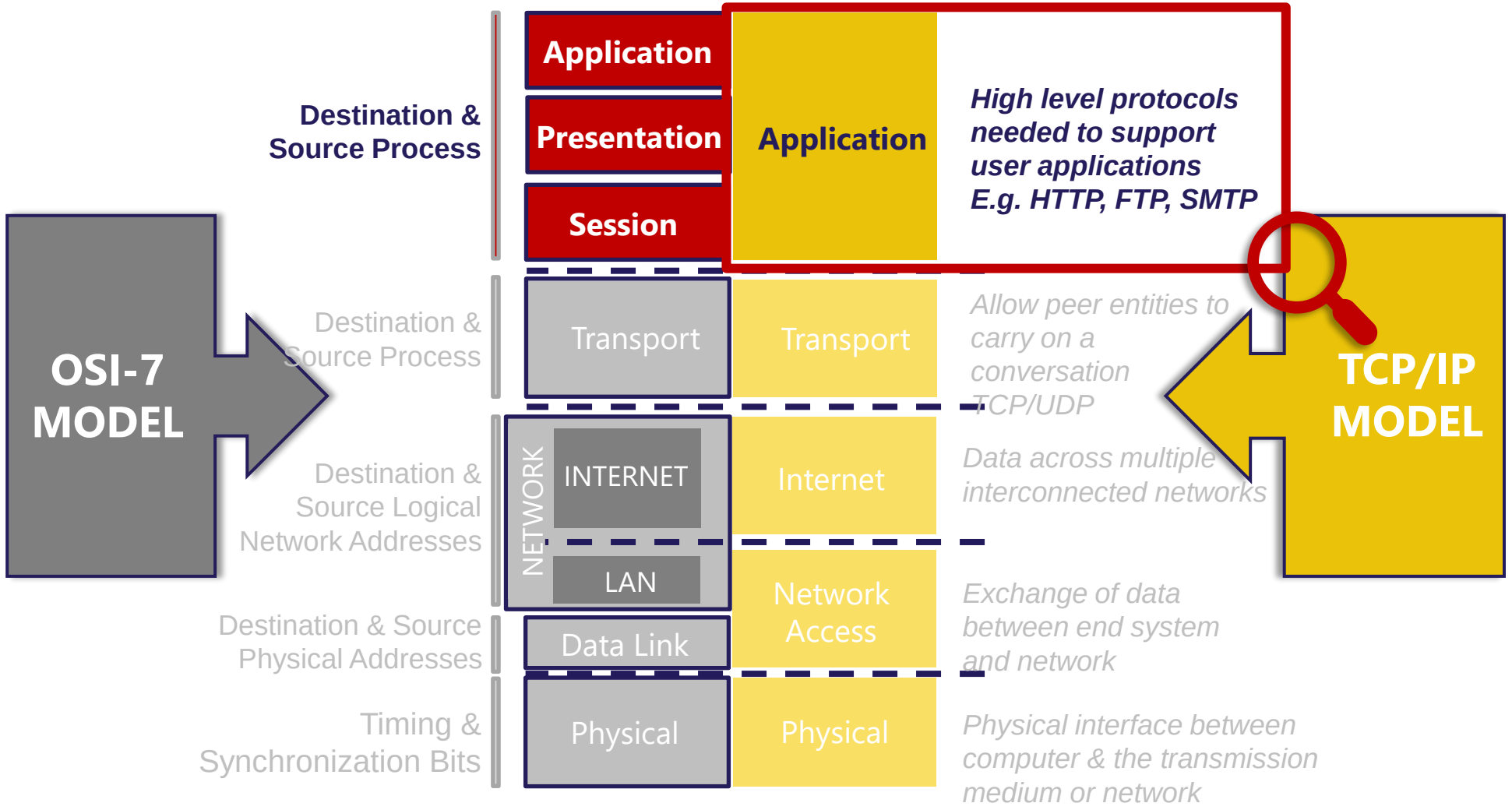
- -Telnet
- -FTP
- -HTTP
- -SMTP/POP3/IMAP

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Application Layer



The application layer specifies communications protocols and interface methods used in process-to-process communications


OSI-7 MODEL
Destination & Source Process
Application
Presentation
Session
Application
High level protocols needed to support user applications E.g. HTTP, FTP, SMTP
Destination & Source Process
Transport
Transport
Allow peer entities to carry on a conversation TCP/UDP
Destination & Source Logical Network Addresses
NETWORK
INTERNET
LAN
Internet
Data across multiple interconnected networks
Destination & Source Physical Addresses
Data Link
Network Access
Exchange of data between end system and network
Timing & Synchronization Bits
Physical
Physical
Physical interface between computer & the transmission medium or network
TCP/IP MODEL

Application Layer

- It takes care of all the communication applications/processes
- Layers below provide reliable transport
- These applications are built on top of the lower layers
- Application layer in TCP/IP covers the Session, Presentation, and Application Layers of the OSI Reference Model

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Some Applications

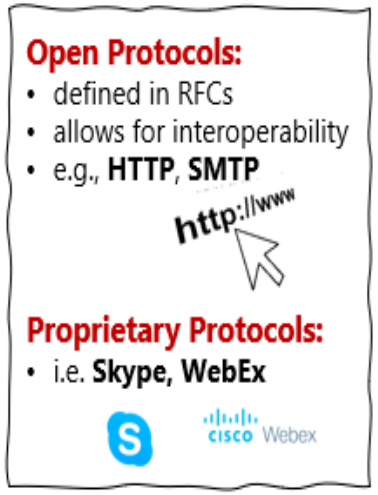
- E-mail
- Web
- Remote login
- P2P file sharing
- Streaming stored video

- Voice over IP e.g. Skype
- Real-time video conferencing
- Social networking
- Search
- Multi-user network games



App-layer Protocol Defines

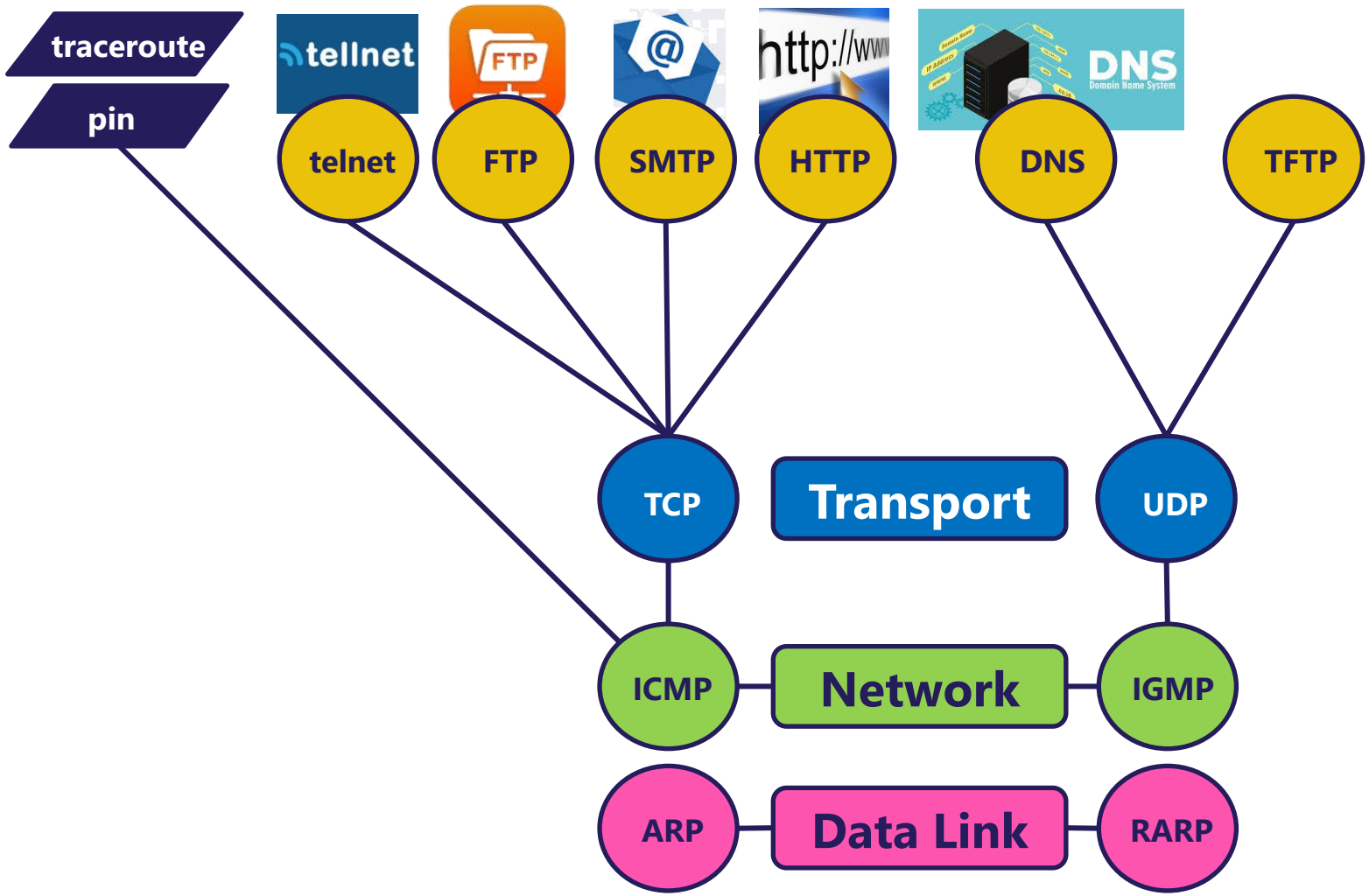
- Message Type *exchanged*
  -e.g., request, response
- Message Syntax
  -What fields in messages & how fields are delineated
- Message Semantics
  -Meaning of information in fields
- Rules for when and how processes send and respond to messages



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Transport Services vs App

Application	Data Loss	Throughput	Time Sensitive
File Transfer	No loss	Elastic	No
E-mail	No loss	Elastic	No
Web Documents	No loss	Elastic	No
Real-time Audio/Video	Loss-tolerant	Audio: 5kbps- 1Mbps Video: 10kbps- 5Mbps	Yes, 100's msec
Stored Audio/Video	Loss-tolerant	Same as above	Yes, few secs
Interactive Games	Loss-tolerant	Few kbps	Yes, 100's msec
Text Messaging	No loss	Elastic	msec (yes and no)


traceroute
pin
telnet
FTP
SMTP
HTTP
DNS
TFTP
TCP
Transport
UDP
ICMP
Network
IGMP
ARP
Data Link
RARP

Secure Communication

TCP & UDP
- -No encryption
- -Cleartext passwords sent into port (socket) traverse the Internet in clear text

SSL (Secure Sockets Layer)
- -Provides encrypted TCP connection
- -Data integrity
- -End-point authentication

SSL is at the app layer
- -Apps use SSL libraries, which “talk” to TCP

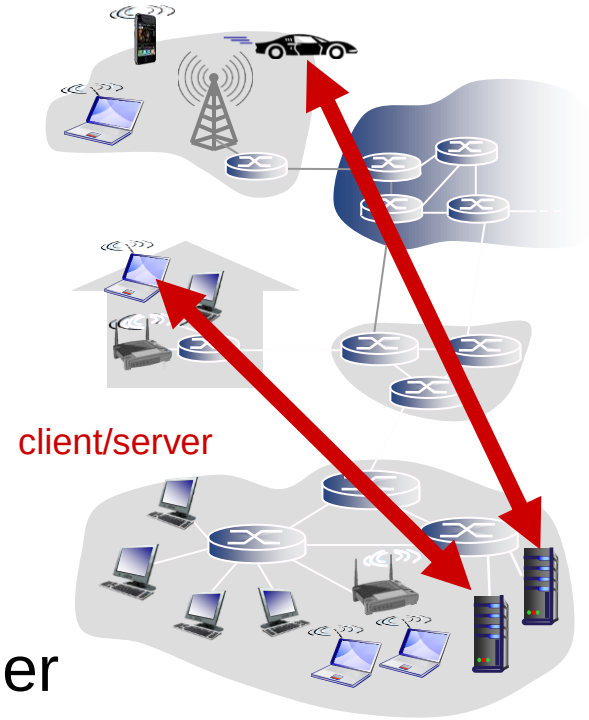


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Client-Server Architecture

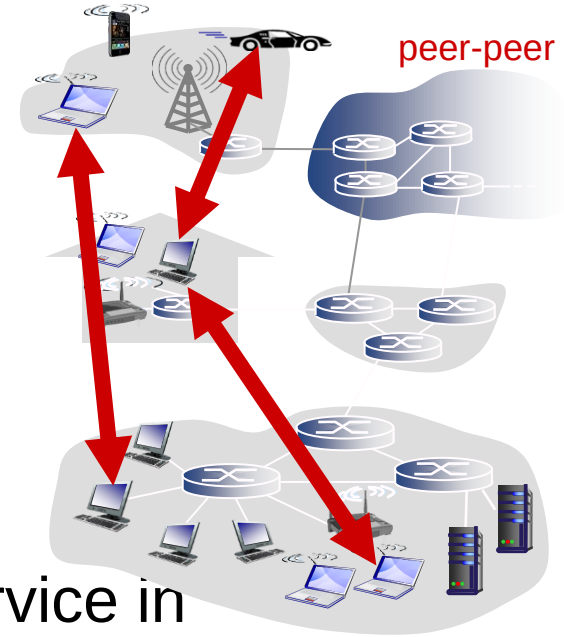
Server:
- Always-on host
- Permanent IP address
- Data centers for scaling

Clients:
- Communicate with server
- May be intermittently connected
- May have dynamic IP addresses
- Do not communicate directly with each other



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P2P Architecture



- Not always on server
- Arbitrary end systems communicate directly
- Peers request service from other peers and provide service in return to other peers
  - Self scalability – new peers bring new service capacity, as well as new service demands
- Peers are intermittently connected and change IP addresses
  - Complex management

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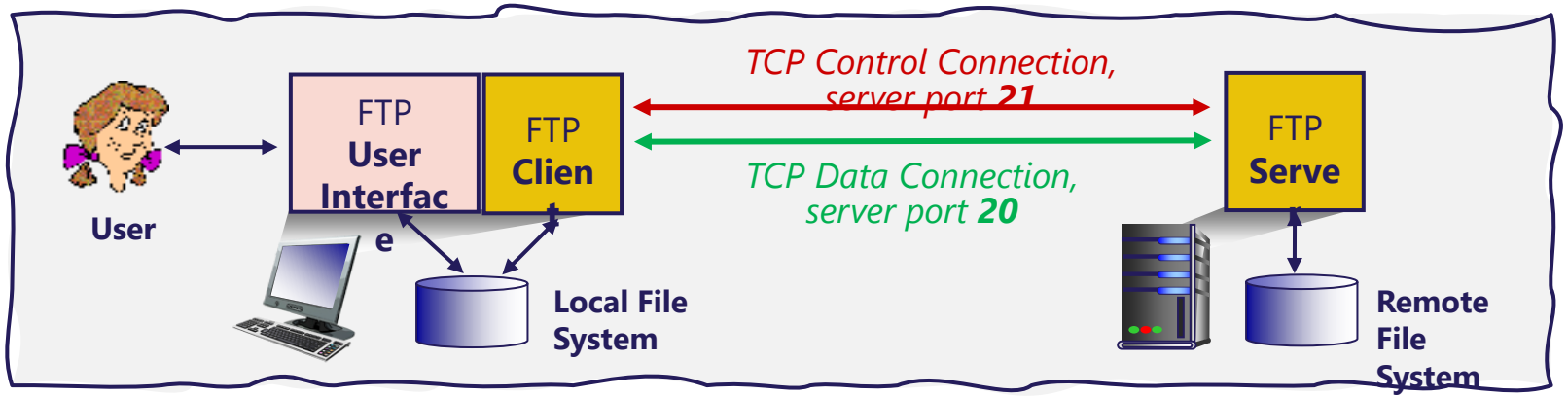
Application Layer Protocols - FTP

- Fundamentals
- Connections
- Commands and Reponses

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File Transfer Protocol (FTP)

- Client initiates connection
- Client authorized over the control connection
- Client browses remote directory, sends commands over the control connection
- When the server gets the file transfer command, the server opens 2nd TCP data connection (for file transfer) to the client



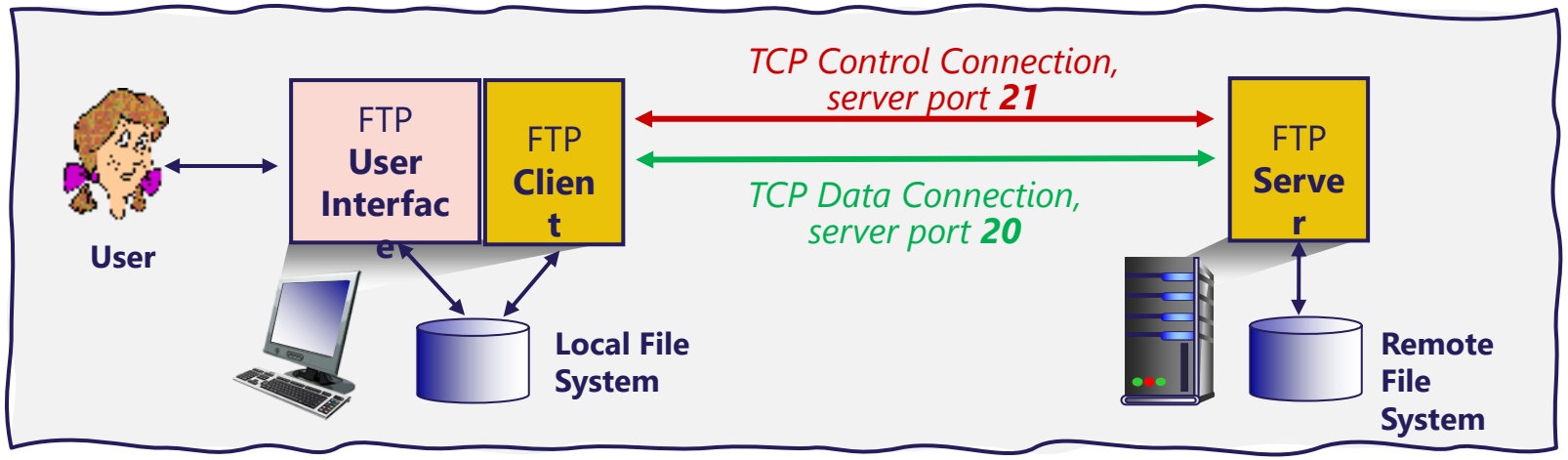
Separate control, data connections

- Control Connection: “out of band”

1 control connection, many data connections (i.e. each file transfer)

- FTP server maintains “state”

  -current directory, earlier authentication, etc.



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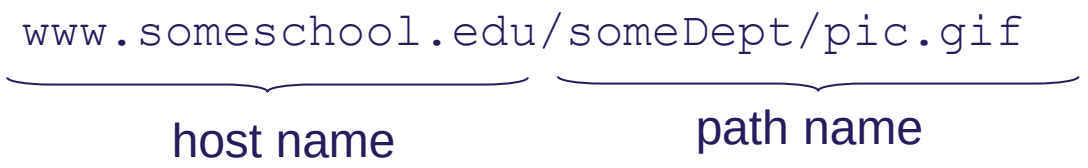
Application Layer Protocols - HTTP

- Review on web
  - -Web resource
  - -WWW
- HTTP
  - -Connections (persistent/non-persistent)
  - -Messages (http request / http response)
  - -HTTP 1.1 / HTTP 2.0
  - -Maintaining State (Cookies)
  - -Web Caching (Browser Cache, Proxy Server)
  - -Conditional GET
  - -Web Sockets

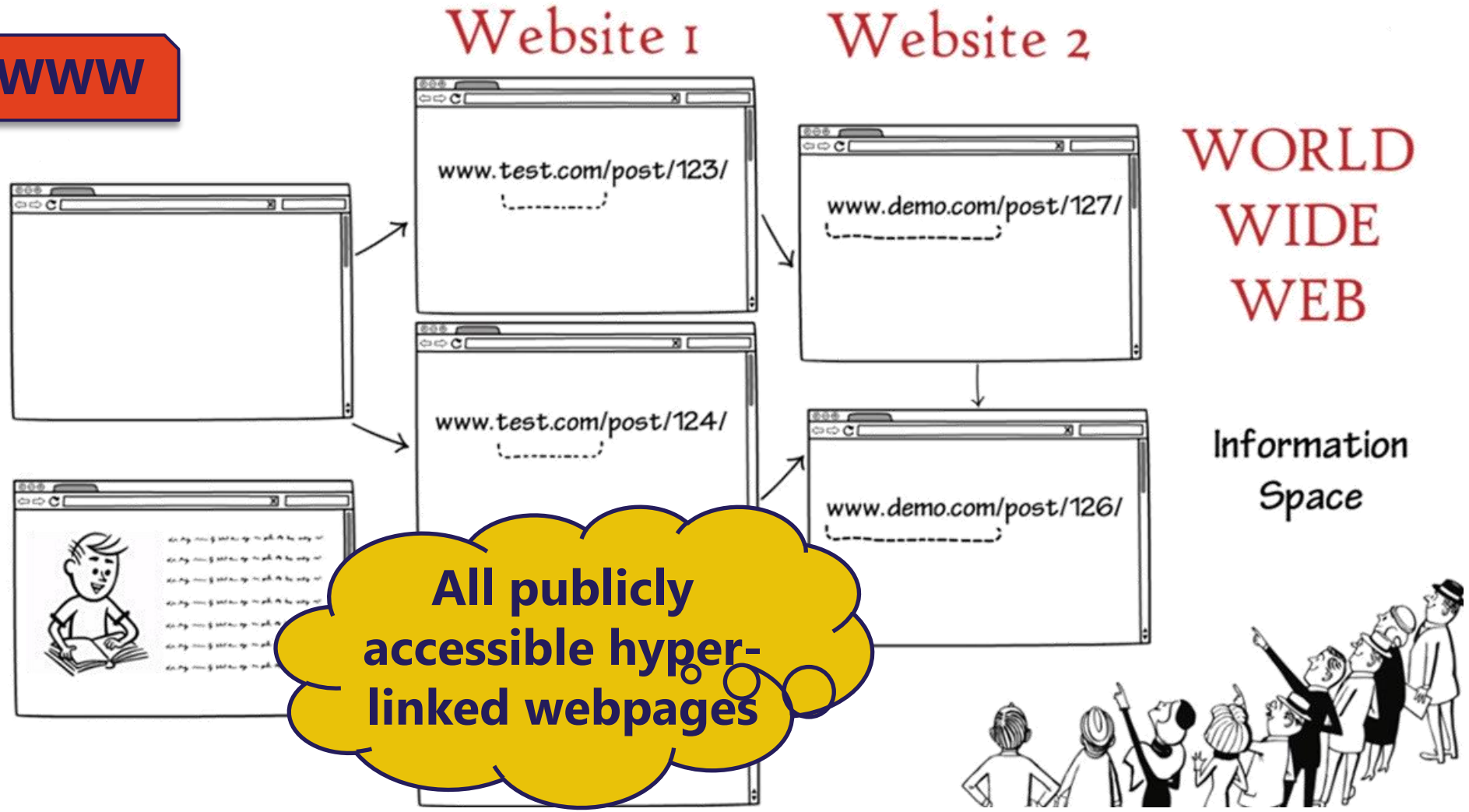
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Web and HTTP

- First, a review…
  - -Web page consists of objects
  - -Object can be HTML file, JPEG image, Java applet, audio file,…
  - -Web page consists of base HTML-file which includes several referenced objects
  - -Each object is addressable by a URL, e.g.



WWW


Website 1
Website 2
www.test.com/post/123/
www.test.com/post/124/
www.demo.com/post/127/
www.demo.com/post/126/
WORLD WIDE WEB
Information Space
All publicly accessible hyper-linked webpages

Web and HTTP – cont.

Is the Internet and World Wide Web the same?

- The Internet is a network of networks

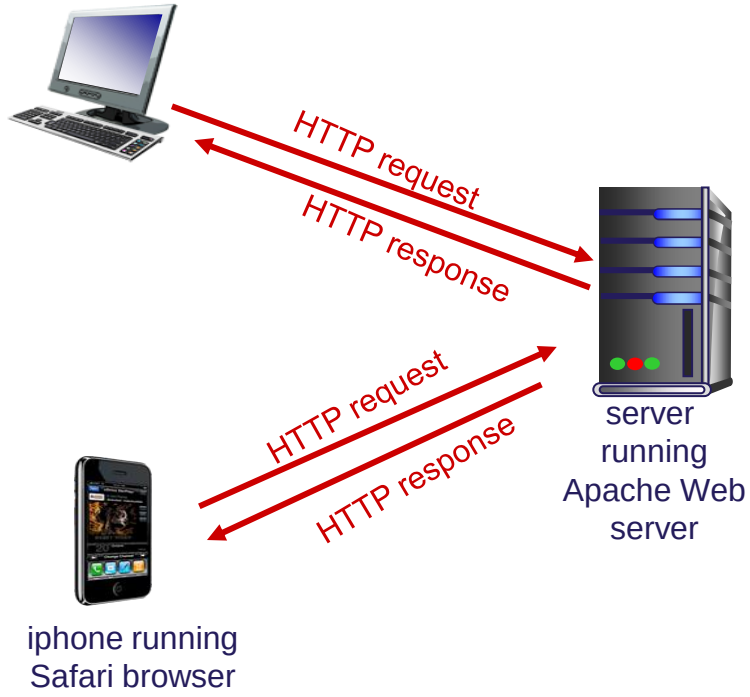
- WWW is a distributed system that runs on top of the Internet

To store information and share it | i.e. website

— Not a network!

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HTTP Overview



- HTTP: hypertext transfer protocol
- Web's application layer protocol
- Client/Server model
  - -Client: browser that requests, receives, (using HTTP protocol) and "displays" Web objects
  - -Server: Web server sends (using HTTP protocol) objects in response to requests

HTTP Overview – cont.

- Uses TCP:
  - •Client initiates TCP connection (creates socket) to the server, port 80
  - •Server accepts TCP connection from client
  - •HTTP messages (application-layer protocol messages) exchanged between browser (HTTP client) and Web server (HTTP server)
  - •TCP connection closed
  - -HTTP is “stateless”
    - • The server maintains no information about past client requests



HTTP Connections

- Non-persistent HTTP
  - -At most one object sent over a TCP connection
  - -Connection then closed
  - -Downloading multiple objects requires multiple connections

- Persistent HTTP
  - -Multiple objects can be sent over a single TCP connection between client and server

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Application Layer Protocols - SMTP, POP3, IMAP

- Mail-sending Protocol
  -SMTP
- Mail-Access Protocol
  -HTTP (Web-mail)
  -POP3
  -IMAP

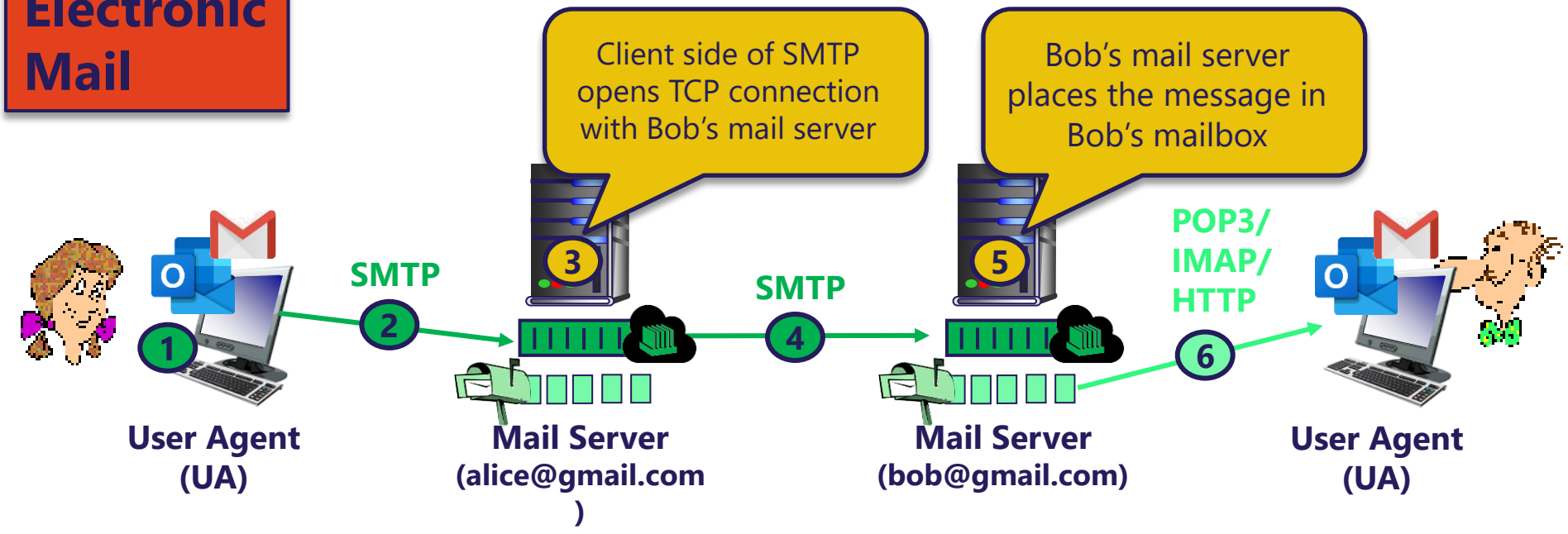
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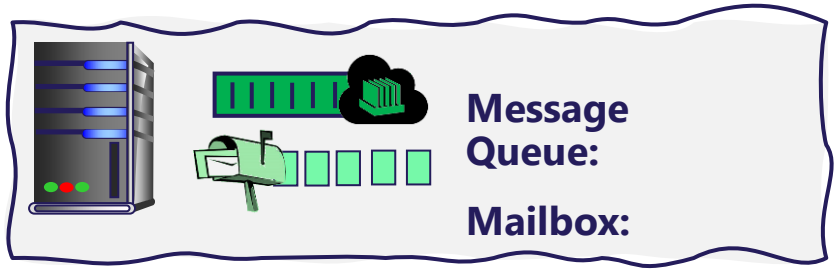
Electronic Mail



1. User Agents
   - Composing, editing, and reading mail messages
   - Outgoing, and incoming messages stored on the server
2. Mail Servers
   - “Mailbox”: incoming messages for user
   - Message Queue: outgoing (to be sent) mail messages
3. SMTP: Simple Mail Transfer Protocol
   - “client”: sending mail server
   - “server”: receiving mail server

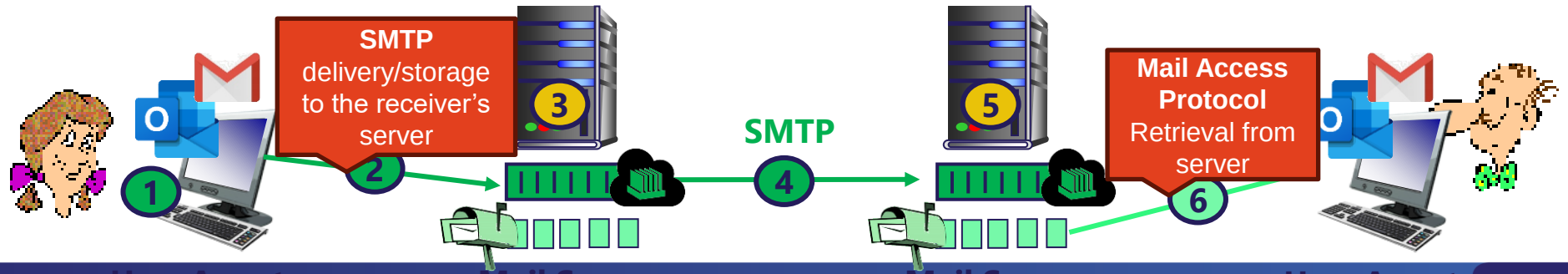
Electronic Mail



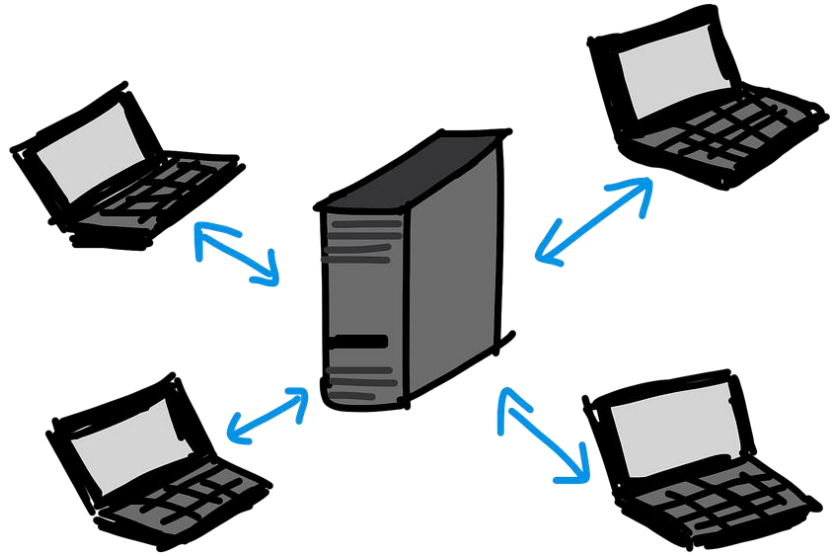


Mail Access Protocols

1. POP: Post Office Protocol: Authorization, Download
2. IMAP: Internet Mail Access Protocol: More Features
   including manipulation of stored messages on the server
3. HTTP: Gmail, Hotmail, Yahoo! Mail, etc.



IMAP



- Keeps all messages in one place: at the server
- Allows user to organize messages in folders
- IMAP is stateful across sessions
  - Names of folders and mappings between message IDs and folder name

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Thank you

